Strategy

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Strategy

Purpose of this chapter

The Strategy chapter translates motivation into what an NSI must be able to do and how it creates value. Motivation explains *why* change is needed. Strategy describes *what* the organisation needs: its capabilities, the resources behind them, the value streams they support and the courses of action that develop them.

These elements are deliberately independent of organisation structure, processes and systems. Departments, processes and applications change often. Capabilities and value streams change slowly. Anchoring architecture work in this layer lets an NSI compare itself with peers, plan investments over several years, and reorganise or replace technology without losing sight of what it must deliver.

Position in the framework

The Strategy layer shares the Strategy & Motivation area of the TOGAF ADM with the Motivation layer (see *Position in the framework* in the Motivation chapter). It receives direction from goals and outcomes. It gives direction to the business architecture: value stream stages are realised by business processes, and capabilities are realised by people, processes, information and technology in the lower layers.

PLACEHOLDER: Position of the Strategy layer between Motivation and the architecture domains

In TOGAF terms, the Strategy chapter supports capability-based planning. Capabilities and value streams are part of the Architecture Vision (Phase A) and of the baseline and target Business Architecture (Phase B). Courses of action and resources return in Phases E and F, where the NSI decides how and in which order capabilities are developed.

How NSIs use this chapter

The capability model and the value streams in this chapter are a reference. Most NSIs perform the same core activities, so these elements are largely shared across the community. An NSI uses them in the same three ways as the motivation elements: adopt, adapt or extend.

The capability model is the most practical starting point. An NSI can assess the maturity of each capability, mark where its goals require improvement, and use the result to prioritise investment. Value streams show where in the production chain a capability creates value, and which stakeholders notice a weak capability first.

Resources and courses of action are specific to each NSI. The SAF explains these concepts and gives examples. The toolkit provides templates to record them.

PLACEHOLDER: Capability maturity assessment and capability heatmap to be added to the toolkit

Strategy concepts in the SAF

SAF section	ArchiMate element	Question it answers	Identifier
Capability Model	Capability	What must the NSI be able to do?	Capability number (e.g. 1.2.1)
Value Streams	Value Stream	Which sequence of stages creates value for a stakeholder?	VS-01 to VS-10
Resources	Resource	Which assets does the NSI have or need to perform its capabilities?	Defined by the NSI
Courses of Action	Course of Action	Which approach will the NSI take to develop its capabilities and reach its goals?	Defined by the NSI

The concepts form a chain. Capabilities are what the NSI can do. Resources are what it uses to do it: people, data, knowledge, technology and money. Value streams show where capabilities create value for a stakeholder. Courses of action describe how the NSI will strengthen its capabilities and resources to reach its goals.

Capability Model

The concept

Within the Statistical Architecture Framework (SAF), business capabilities define the fundamental abilities of a statistical organization to perform its mission , independent of processes, organizational structures, or technologies. A business capability represents what the organization must be able to do to deliver value and achieve its objectives, providing a stable and long-term view of the enterprise's functional landscape.

Capabilities serve as the enabling building blocks of the statistical organization. They do not change frequently, even when the underlying processes, applications, or organizational units evolve. For example, the capability "Manage Statistical Data Quality" remains essential regardless of whether data validation is manual, automated, or AI-driven. This stability makes capabilities the preferred unit of analysis for strategic planning, transformation roadmaps, and architecture alignment within the SAF.

Three characteristics set capabilities apart:

- **They are stable.** Statistical quality management (2.1.3) remains essential whether quality checks are manual, automated or AI-supported. Processes and applications change; the capability stays.
- **They are independent of organisation.** A capability can be spread over several departments, or one department can hold several capabilities. The capability model does not follow the organisation chart.
- **They are hierarchical.** High-level capabilities are decomposed into more detailed ones. In the SAF, *Data management* (2.1) is decomposed into, among others, *Metadata management* (2.1.2).

Each capability describes a specific business purpose and outcome, which is realized through one or more business functions and operationalized by business processes. Capabilities therefore serve as the conceptual bridge between strategy and execution. They translate strategic goals into operational competence. Within the SAF, capabilities are often organized hierarchically, showing decomposition from high-level domains (such as Data Governance or Statistical Production) into more granular sub-capabilities (such as Metadata Management or Sampling Design).

How the SAF applies capabilities

Why the SAF uses a capability model

Capabilities are the preferred unit for strategic planning in the SAF. Because they are stable, they make a reliable basis for multi-annual roadmaps. Because they are independent of organisation and technology, they let NSIs compare themselves with each other. A capability view also makes gaps visible: where a goal requires a capability that is weak or missing, investment is needed.

How capabilities fit into an NSI's own architecture

An NSI maps the SAF capability model onto its own organisation. It checks which capabilities it has, which it lacks and which are specific to its mandate. It then assesses the maturity of each relevant capability and links capabilities to its goals and outcomes. Capabilities with a low maturity and a high contribution to goals become candidates for investment. Each capability can then be traced to the processes, applications and platforms that realise it.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Goals and outcomes (Motivation)	Capabilities realise goals and outcomes	Metadata management (2.1.2) realises enhanced data quality (SAF-O2)
Assessment (Motivation)	A capability maturity assessment is a structured form of assessment	A low maturity score for Statistical quality management (2.1.3)
Value Streams	Capabilities enable value stream stages	Primary data collection (1.2.1) enables stage *Collect or Receive Data* of VS-01

Related concept	Relationship	Example in the SAF
Resources	Capabilities are performed with resources	Methodology (2.2.3) depends on staff with methodological expertise
Courses of Action	Courses of action develop capabilities	A shared platform with partner NSIs develops IT infrastructure (3.2.5)
Business functions and processes (Business Architecture)	Business functions and processes realise capabilities	Business function *Classify Data* realises Data standardisation (1.3.1)
Steady states (Business Architecture)	Capabilities produce or govern steady states	Data standardisation (1.3.1) produces State #2 Standardized Data

Capability map

The SAF capability model has four main levels.

PLACEHOLDER: SAF capability map (ArchiMate)

ID	Main level	Description	Reference
1	Primary statistical capabilities	The ability to perform the statutory statistical tasks of an NSI.	GAMSO
2	Supporting data capabilities	The ability to manage statistical data and to carry out statistical research.	
3	Supporting business capabilities	The ability to make sure that the right people, information and resources are available to support the statistical and governance capabilities.	
4	Governance capabilities	The ability to give direction to the organisation, by setting its strategy and the framework within which change takes place.	GAMSO 1.2, Strategy and Leadership

Capability catalogue

1. Primary statistical capabilities

ID	Capability	Description	Reference
1.1	**Statistical design and build**	The ability to identify information needs and set up statistical processes to meet them.	GSBPM 5.2, Design

ID	Capability	Description	Reference
1.1.1	Needs inventory	The ability to identify the demand for statistical information and translate it into the societal questions the NSI will respond to. Needs are often identified through strategic relationships with users and stakeholders. The result is usually laid down in the NSI's multi-annual programme and the process that prepares it.	GSBPM
1.1.2	Design statistics product	The ability to translate information needs into concrete designs of datasets, process and workflow models, and rule sets and methodology, for the datasets that are released. This covers both the statistical product and the process that produces it. The process consists of process steps, rule sets for each step, the methodology, the process flow, the designs of the required datasets and the designs of shared connection points for reuse. If a required dataset is not available, a data collection assignment can follow. For primary data collection, designs then follow for sampling, questionnaires and similar instruments.	GSBPM, Design phase
1.1.3	Build statistics product	The ability to build, test and manage a statistical product until the end of its life cycle. The outputs of the design are assembled and configured into a complete operational environment in which the process can run.	GSBPM
1.2	**Data collection**	The ability to collect data for official statistics.	GSBPM 5.2, Collect

ID	Capability	Description	Reference
1.2.1	Primary data collection	The ability to set up and manage surveys up to completion, and to transfer the collected data and metadata for processing and analysis. Primary data collection uses one or more collection modes. The target population and sample size are set together with the client, based on the information need. Person surveys and business surveys are distinguished. A survey consists of, among other things, a sample, a questionnaire and an approach strategy.	ESS capability model
1.2.2	Secondary data collection	The ability to collect data and metadata from third parties for the production of statistics. Data that third parties have already collected is used as much as possible.	ESS capability model
1.3	**Processing**	The ability to process raw data into statistical microdata and statistics.	GSBPM, Process phase
1.3.1	Data standardisation	The ability to standardise data without loss of content. Standardisation covers data formats, category codes, linkage keys and units of measurement, such as weight, distance and time. It takes place directly after the raw data is received. Standardising as early as possible makes the rest of the process more efficient.	

ID	Capability	Description	Reference
1.3.2	Deriving variables and units	The ability to create new units, and values of new variables, from existing units and values of existing variables. New variables are usually derived to describe reality better. The data source was often not designed for that purpose.	
1.3.3	Check content	The ability to check datasets for irregularities, such as missing data, possible errors and outliers. Detected errors can lead to adjustment (see 1.3.4). They can also lead to a quality indicator that is used in quality reporting. Detected outliers lead to an outlier indicator, which matters for a good estimation method (see 1.3.5).	GSBPM
1.3.4	Adjusting and imputing	The ability to adjust incorrect values and to fill in missing values. Together with *Check content* (1.3.3), this capability is often called editing. Checks use editing rules, and adjustment is often based on imputation. Imputed values do not yet have to meet the editing rules. A final adjustment therefore often follows, with the editing rules as constraints.	GSBPM, ESS capability model

ID	Capability	Description	Reference
1.3.5	Estimating population parameters	The ability to estimate population parameters from microdata. Estimated population parameters often appear as observations, the cells, in a dimensional dataset such as a table. There are several estimation methods, including the traditional "weight and aggregate". Weights can be derived with several techniques. Sometimes weights are adjusted to give detected outliers less influence.	
1.3.6	Integrate statistics	The ability to integrate separate statistics into one consistent whole. Examples are integration systems such as the national accounts, sector accounts and labour accounts.	
1.3.7	Statistical disclosure	The ability to check datasets for privacy sensitivity and to adjust them where confidentiality would be violated.	GSBPM 6.4
1.4	**Analysis**	The ability to examine and understand data before it is disseminated. In NSIs, statistical analysis is also called statistical research. It draws connections or conclusions from relationships in the data, such as correlations.	GSBPM 5.2, Analyse
1.4.1	Statistical (output) analysis	The ability to validate, interpret and explain statistical output, and to finalise it for dissemination using shared methods and processes.	ESS capability model
1.5	**Dissemination**	The ability to manage the release of statistical products to internal and external users.	GSBPM 5.2, Disseminate

ID	Capability	Description	Reference
1.5.1	Publication management	The ability to manage the release of statistical output and related content according to release calendars, so that users have predictable and equal access.	ESS capability model
1.5.2	Promotion to use	The ability to promote statistical output to potential users, and to inform the press and other interested parties about it.	ESS capability model
1.5.3	Facilitate data access	The ability to make statistical data and metadata available to people and machines through multiple channels.	ESS capability model
1.5.4	Statistical information management	The ability to prepare and distribute publications about statistical output, such as press releases, interpretations and reports.	ESS capability model

2. Supporting data capabilities

ID	Capability	Description	Reference
2.1	**Data management**	The ability to manage data and metadata.	
2.1.1	Data lifecycle management	The ability to manage data throughout its life cycle.	
2.1.2	Metadata management	The ability to manage metadata throughout its life cycle.	ESS capability model
2.1.3	Statistical quality management	The ability to assess quality, put control mechanisms in place and start quality improvements across the statistical value chain.	
2.1.4	Statistical register management	The ability to set up, maintain and provide register services for collecting and integrating data. This includes the management of master data.	ESS capability model

ID	Capability	Description	Reference
2.2	**Research**	The ability to build and record knowledge about statistical methods, products, sources and techniques.	
2.2.1	Data source exploration	The ability to explore new data sources.	ESS capability model
2.2.2	Product innovation	The ability to create new statistical products from existing data sources, and to explore new data sources that are useful and important to users. Product innovation can precede the design of a statistical product (see 1.1.2).	ESS capability model
2.2.3	Methodology	The ability to investigate new methods for the statistical process.	
2.2.4	Data services	The ability to provide data for statistical or scientific research. This applies to both internal and external datasets. These services are provided through, among other things, microdata services.	
2.2.5	User research	The ability to collect and assess user needs, and translate them into statistical output.	

3. Supporting business capabilities

ID	Capability	Description	Reference
3.1	**Business operations**	The ability to manage personnel, purchasing and contracts, finance and housing, including the related systems and communication.	
3.1.1	Personnel management (HR)	The ability to make sure that enough qualified staff are available to carry out all business processes.	
3.1.2	Purchasing and contract management	The ability to acquire goods and services, and to monitor the agreements made with suppliers.	

ID	Capability	Description	Reference
3.1.3	Financial management	The ability to plan, organise, design, record and control the financial activities of the organisation.	
3.1.4	Communication management	The ability to communicate correctly with internal and external parties.	
3.1.5	Administrative support	The ability to support administrative tasks.	
3.1.6	Facility management	The ability to integrate people, place and process within the built environment. The description is based on ISO 41011. Facility management aims to improve the quality of life of people and the productivity of the core business.	ISO 41011
3.1.7	Legal support	The ability to handle requests and applications, assess agreements and advise on relevant laws and regulations.	
3.1.8	Information management and archive management	The ability to keep the organisation's information sustainably accessible (FAIR). This concerns information created in supporting and management processes, not statistical information from the primary statistical process.	
3.2	**Computerisation and automation**	The ability to translate the organisation's information needs into functionality of the information provision, and to keep information systems available to support business processes.	
3.2.1	Application development	The ability to develop applications in-house. This capability is needed to fill gaps in functionality that the statistical process requires.	

ID	Capability	Description	Reference
3.2.2	Functional management	The ability to manage applications so that they support business processes as well as possible. An example is collecting user requirements and translating them into new application functionality.	
3.2.3	Technical management	The ability to manage system software and applications. This covers installing, configuring and managing hardware and software so that it keeps working correctly, including all measures to resolve disruptions in time.	
3.2.4	Digital workplace management	The ability to manage the resources and services that staff use in their digital workplace. Examples are laptops and desktops, mobile devices, virtual desktops, video and audio conferencing, printing, VPN and proxy services.	
3.2.5	IT infrastructure	The ability to develop, maintain and manage IT infrastructure.	
3.3	**Security and privacy**	The ability to carry out all security and privacy activities.	
3.3.1	Privacy management [**S7**]	The ability to develop, manage, apply and test the NSI's privacy policy, in order to comply with privacy laws and regulations.	
3.3.2	Security planning	The ability to identify, analyse and plan measures to secure systems, networks, the organisation and its facilities.	
3.3.3	Security incident management	The ability to identify, analyse, respond to and recover from security incidents.	

ID	Capability	Description	Reference
3.3.4	Logical identity and access control	The ability to make sure that a person, organisation or IT facility can only use the automated functions for which access rights have been granted.	
3.3.5	Physical security and access control	The ability to provide controlled and safe access to buildings and spaces, and to keep the people and goods in them safe.	
3.3.6	IT security services	The ability to provide IT services that support information security. Examples are endpoint protection, PKI and network security management.	
3.4	**Customer and chain interaction**	The ability to interact with customers and chain partners: providing them with the right data and information, and receiving notifications, requests and data.	
3.4.1	Services management	The ability to design, develop, implement and manage the services that the organisation provides to customers and chain partners.	
3.4.2	Contact management	The ability to maintain relationships with customers and chain partners.	GAMSO 1.2, Manage Consumers **[S9]**
3.4.3	Service agreement	The ability to draw up, manage and enforce service agreements with clients and chain partners. Service agreements apply to clients that receive services from the NSI, such as access for external researchers or additional statistical services. They also apply to parties that supply data for primary or secondary data collection. If a party does not comply, the NSI can take enforcement action.	

ID	Capability	Description	Reference
3.4.4	Self-service	The ability to let customers and chain partners answer questions about the services they use, or make justified changes to them, without help from staff.	

4. Governance capabilities

ID	Capability	Description	Reference
4.1	**Strategy and leadership**	The ability to set the direction of the organisation and to manage change towards it.	GAMSO 1.2, Strategy and Leadership
4.1.1	Strategy and governance	The ability to develop a vision, and to set up and monitor the organisation and its management.	
4.1.2	Policy and planning	The ability to translate the strategy into concrete objectives, principles and plans.	
4.1.3	Change management	The ability to define and manage major changes, so that they contribute as much as possible to the objectives.	
4.1.4	Improvement management	The ability to manage daily operations, identify areas for improvement and implement improvements based on daily operations.	
4.1.5	Accountability	The ability to report to stakeholders inside and outside the organisation on how far obligations and agreements are met.	

ID	Capability	Description	Reference
4.1.6	Collaboration	The ability to maintain and strengthen strategic relationships with external stakeholders. Collaboration covers coordination with other statistical organisations and other external stakeholders. Within a statistical system, this coordination can follow a geographical hierarchy (local, regional, national, multinational) or a division of responsibilities between organisations. It includes identifying new opportunities for data exchange and integration. It also gives the statistical community opportunities to exchange knowledge, improve statistical infrastructure and practice, and influence statistical standards.	GAMSO 1.2, Manage Strategic Collaboration and Cooperation
4.1.7	Architecture Management	The ability to develop, maintain and govern the enterprise architecture of the NSI, and to use it in decisions on strategy, investment and change.	

PLACEHOLDER: Capability Model Template and the application of Capabilities to be added to the toolkit

Value Streams [T1]

The concept

A value stream is a sequence of stages that together create an overall result for a stakeholder. Each stage adds a specific piece of value. The value stream ends when the stakeholder receives the result.

Three characteristics set value streams apart:

- **They are seen from the stakeholder.** A value stream starts with a need and ends with value for someone. It does not follow the organisation chart or the order of systems.
- **They describe value, not work.** A stage says *what value* is added. *How* the work is done is described by business processes in the Business Architecture.
- **They are enabled by capabilities.** Each stage needs one or more capabilities. A weak capability shows up as a weak stage.

Value streams and capabilities are two views on the same organisation. Capabilities show *what* the NSI can do. Value streams show *where* that ability turns into value.

How the SAF applies value streams

Why the SAF uses value streams

Value streams connect the capability model to the stakeholders. They make clear which capabilities matter for which result, and which stakeholders notice a weak capability first. They also give the Business Architecture its structure: each stage is realised by business processes, and the primary value streams follow the steady states of data.

How value streams fit into an NSI's own architecture

Most NSIs produce statistics in a similar sequence, so the SAF value streams fit most of them. An NSI checks whether the stages match its own production chain, adapts names where needed, and links its own capabilities to each stage. It can then use the value streams to assess where in the chain improvements create the most value for users.

Relationships with other concepts

Related concept	Relationship	Example in the SAF
Stakeholders (Motivation)	A value stream creates value for a stakeholder	VS-06 creates value for users of statistics (SAF-ST02, SAF-ST03, SAF-ST06)
Value (Motivation)	A value stream delivers value elements	VS-06 contributes to trust and legitimacy (SAF-V6)
Capability Model	Capabilities enable value stream stages	Primary data collection (1.2.1) enables VS-01.3 *Collect or Receive Data*
Courses of Action	Courses of action improve value streams by developing capabilities	A web-first collection strategy improves VS-01
Steady states (Business Architecture)	Each primary value stream produces one steady state	VS-02 produces State #2 Standardized Data
Business processes (Business Architecture)	Business processes realise value stream stages	The execution processes of the Standardization domain realise the stages of VS-02

The processes that realise each stage, and the steady states between them, are described in the Business Architecture chapter.

Value stream overview [T2] [T3]

The SAF has ten value streams. Six primary value streams form the statistical value chain. Four supporting value streams run across the chain.

ID	Value stream	Type	Receives value	Value created	Steady state in → out
VS-01	Data Acquisition and Ingestion	Primary	Statistical production; data providers	Trust and readiness: data acquired correctly and ready for use	#0 → #1
VS-02	Data Standard-ization and Integration	Primary	Statistical production; reusers of microdata	Coherence and comparability	#1 → #2

ID	Value stream	Type	Receives value	Value created	Steady state in → out
VS-03	Data Editing and Enrichment	Primary	Statistical production; methodologists	Reliability and analytical readiness	#2 → #3
VS-04	Estimation, Analysis and Statistical Modeling	Primary	Statistical product owners	Analytical insight and statistical credibility	#3 → #4
VS-05	Statistical Product Composition	Primary	Statistical product owners; users	Coherence and interpretability	#4 → #4 (approved)
VS-06	Dissemination and Service Delivery	Primary	Users: policy makers, researchers, media, businesses, the public	Impact and accessibility	#4 → #5
VS-07	Governance and Compliance	Supporting	Management, supervisory bodies, society	Integrity and accountability	Across all states
VS-08	Innovation and Methodological Development	Supporting	Statistical production	Continuous improvement	Across all states
VS-09	Infrastructure and Platform Enablement	Supporting	All value streams	Operational capability and continuity	Across all states
VS-10	Customer and Stakeholder Engagement	Supporting	Users, data providers, partners	Relevance and trust	#5 → (#6), feedback to VS-01

Capabilities per value stream [T4]

PLACEHOLDER: ArchiMate view of capabilities serving value stream stages

Primary Value Streams

VS-01 Data Acquisition and Ingestion

The Data Acquisition & Ingestion Value Stream is the first step in the statistical value chain. It focuses on bringing raw data into the statistical system in a secure, traceable, and compliant way. The purpose of this value stream is to make sure that data, no matter where the data comes from, enters the organization in a state that is authentic, well-documented, and ready for use. This value stream sets the foundation for everything that follows. It makes sure that the right data is collected, that legal requirements and ethical guidelines are observed, and that the data is safely stored and traceable from its source.

PLACEHOLDER: Add ArchiMate view of VS-01

Stage	Name	Value added
VS-01.1	Identify Data Needs and Sources	The information need is clear, and suitable sources are known
VS-01.2	Establish Data Agreements and Access Rights	Data can be obtained lawfully, under agreed conditions

Stage	Name	Value added
VS-01.3	Collect or Receive Data	Data and its metadata are in NSI custody
VS-01.4	Validate and Log Data Intake	The delivery is complete, intact and traceable
VS-01.5	Store and Register Raw Data Artefacts	Raw data is registered and available for reuse (State #1)

The value created here lies in trust and readiness , trust that the data has been acquired correctly, and readiness for the data to move forward into later stages of the statistical process.

Once data has been safely acquired and registered, it is ready to move into the next phase - the Data Standardization & Integration Value Stream. Here, the focus shifts from collecting data to structuring it, ensuring that all incoming data can be compared, combined, and analyzed.

VS-02 Data Standardization and Integration

The Data Standardization & Integration Value Stream is the second step in the statistical value chain. After raw data has been collected and securely stored, it now needs to be standardized, harmonized, and prepared for further processing. In this value stream, the organization ensures that data from different sources - surveys, administrative registers, or external datasets - can work together seamlessly. This is achieved through consistent structures, common formats, and shared definitions. The main goal is interoperability: data that looks different at the source must be aligned so it can be understood, compared, and reused across systems, domains, and time periods.

PLACEHOLDER: Add ArchiMate view of VS-02

Stage	Name	Value added
VS-02.1	Profile and Assess Data Structure	Differences from the target model are known
VS-02.2	Apply Standard Classifications and Formats	Data uses common codes, formats and identifiers
VS-02.3	Align Metadata and Semantics	Definitions and descriptions are consistent
VS-02.4	Validate Structural and Semantic Conformance	Conformance to standards is confirmed
VS-02.5	Register Standardized Artefacts	Standardized data is registered and available for reuse (State #2)

The value created here is coherence and comparability, transforming raw data into a structured and standardized asset that can flow smoothly into editing, analysis, and modeling.

Once data is standardized and validated, it can safely move to the Data Editing & Enrichment Value Stream. In that stage, attention shifts from structure and standards to quality and completeness. Here the organization begins improving data accuracy, filling gaps, and enriching datasets with derived or corrected information.

VS-03 Data Editing and Enrichment

The Data Editing & Enrichment Value Stream is the third step in the statistical value chain. After data has been standardized and aligned, the next task is to improve its quality, completeness, and usability. At this stage, the organization focuses on cleaning, correcting, and enriching data so that it becomes trustworthy and analytically sound. Errors are detected, inconsistencies are corrected, missing values are filled, and new variables are derived to add more meaning and analytical value. This is where raw, standardized data begins to transform into high-quality statistical input , ready for estimation, modeling, and analysis.

PLACEHOLDER: Add ArchiMate view of VS-03

Stage	Name	Value added
VS-03.1	Conduct Error Detection and Quality Diagnostics	Errors, duplicates, gaps and outliers are known
VS-03.2	Perform Data Editing and Correction	Incorrect values are corrected
VS-03.3	Impute Missing or Non-Response Data	Missing values are filled with documented methods
VS-03.4	Derive Additional Variables and Indicators	Data describes reality better than the source alone
VS-03.5	Perform Quality Evaluation and Versioning	Quality is documented, and processed microdata is registered (State #3)

The main value created here is data reliability and analytical readiness. By systematically improving data quality, the organization ensures that every subsequent result , every estimate, model, and publication, rests on a solid foundation.

VS-04 Estimation, Analysis and Statistical Modeling

The Estimation & Analysis Value Stream is the fourth stage of the statistical value chain. Here, the focus shifts from improving data quality to extracting insight and meaning. At this stage, the organization applies statistical methods, estimation techniques, and analytical models to transform edited and enriched data into representative results that describe real-world phenomena. This is where data becomes statistical information, summarized, interpreted, and validated to reflect populations, trends, and behaviors. Every decision taken in this stage must balance methodological detail with transparency and reproducibility.

PLACEHOLDER: Add ArchiMate view of VS-04

Stage	Name	Value added
VS-04.1	Confirm Estimation and Analytical Approach	The method fits the design and the purpose
VS-04.2	Apply Weighting and Calibration	Bias from sampling, non-response and coverage is reduced
VS-04.3	Estimate and Aggregate Data	Statistics and indicators are available
VS-04.4	Perform Model-Based Analysis	Model-based results complement direct estimates
VS-04.5	Validate and Document Results	Results are sound and documented (State #4)

The main value created here is analytical insight and statistical credibility. Through careful estimation and analysis, data is turned into knowledge that policymakers, researchers, and society can trust.

After estimation and analysis, the organization moves to the Statistical Product Composition Value Stream. Here, the focus shifts from individual analyses to building integrated and composite products. In that next stage, data from multiple sources and analyses are combined into coherent statistical publications, dashboards, and reports, turning analytical results into accessible products that provide a complete picture.

VS-05 Statistical Product Composition

The Statistical Product Composition Value Stream is the fifth stage in the statistical value chain. After estimation and analysis, the focus now shifts to bringing multiple analytical results together to create

coherent, meaningful, and high-value statistical products. At this stage, data from different sources, domains, and analyses are combined, refined, and formatted into composite datasets, time series, and indicators that can be used for reporting, policymaking, and research. This value stream transforms intermediate analytical outputs into final statistical products , ready for validation, interpretation, and dissemination.

PLACEHOLDER: Add ArchiMate view of VS-05

Stage	Name	Value added
VS-05.1	Select and Integrate Source Datasets	Relevant results are brought together
VS-05.2	Construct Composite Indicators or Indices	Indices, ratios and aggregates are available
VS-05.3	Perform Consistency and Coherence Checks	The product is internally consistent
VS-05.4	Generate Analytical and Interpretive Outputs	Results are explained and visualised
VS-05.5	Approve and Register Statistical Products	The product is approved for release (State #4, approved)

The key value created here is coherence and interpretability. Through careful composition, integration, and quality control, statistical outputs become consistent, accessible, and ready to serve their intended purpose.

After products are approved and registered, they move into the Dissemination & Service Delivery Value Stream. Here, the focus shifts from producing statistics to sharing them, making them available through publications, APIs, dashboards, and data services. That next stage ensures that the work done throughout the entire value chain ultimately reaches users in a secure, accessible, and user-friendly way.

VS-06 Dissemination and Service Delivery

The Dissemination & Service Delivery Value Stream is the final stage in the statistical value chain. At this point, the statistical products created in the previous stages are published, distributed, and made accessible to users through trusted and secure channels. The main goal of this value stream is to ensure that information reaches users in a way that is accurate, timely, and easy to understand. This includes policymakers, researchers, journalists, businesses, and the general public. Dissemination is not only about releasing data, it is about providing services that help users access, interpret, and apply statistical information responsibly. This step transforms statistical outputs into societal value by enabling informed decision-making, evidence-based policy, and transparency.

PLACEHOLDER: Add ArchiMate view of VS-06

Stage	Name	Value added
VS-06.1	Plan Dissemination and Access Strategy	Target users, channels and formats are set
VS-06.2	Prepare Data for Dissemination	Output is disclosure-safe and in the right format
VS-06.3	Publish and Provide Access Services	Users can find and access the output (State #5)
VS-06.4	Support and Engage Users	Users understand and use the output correctly
VS-06.5	Monitor Usage and Feedback	The NSI knows how its output is used

The key value created here is impact and accessibility, ensuring that statistics and data services are usable, trusted, and beneficial to all. The Dissemination & Service Delivery Value Stream relies heavily on several cross-cutting value streams:

- Governance and Compliance ensures that all dissemination complies with data protection and confidentiality laws.
- Infrastructure and Platform Enablement provides the technical platforms for portals, Customer and Stakeholder Engagement ensures that dissemination services meet real user needs and expectations.

Together, these supporting streams ensure that dissemination is not just a technical process , but a strategic, user-centered activity that delivers societal value through trustworthy data.

Supporting value streams

VS-07 Governance and Compliance

The Governance and Compliance Value Stream runs horizontally across all stages of the statistical value chain. Its purpose is to make sure that every process, dataset, and product within the organization operates under clear rules, standards, and accountability mechanisms. This value stream builds trust , both inside and outside the organization , by ensuring that all data activities are ethical, transparent, and compliant with legal, methodological, and quality requirements. It provides the framework of control and assurance that keeps the statistical system reliable, consistent, and aligned with national and international standards such as the European Statistics Code of Practice, GDPR, GSBPM, GAMSO and ISO data quality principles.

PLACEHOLDER: Add ArchiMate view of VS-07

Stage	Name	Value added
VS-07.1	Define Governance Frameworks and Policies	Rules for data, metadata, ethics, confidentiality and quality are set
VS-07.2	Establish Legal and Regulatory Compliance Mechanisms	Applicable rules are known and translated into procedures
VS-07.3	Implement Data and Metadata Governance	Ownership, stewardship and lineage are assigned
VS-07.4	Monitor Quality and Compliance Indicators	Quality and compliance status is visible
VS-07.5	Audit, Certify and Report Compliance	Compliance is demonstrated
VS-07.6	Update and Improve Governance Mechanisms	Governance keeps up with new rules and lessons learned

The value created here is integrity and accountability making sure that the NSI acts responsibly, protects privacy, and delivers products users can trust.

The Governance and Compliance Value Stream supports all other value streams by acting as the foundation of trust and assurance. It ensures that Data Acquisition & Ingestion follows lawful and ethical standards. It guarantees that Editing, Estimation, and Dissemination meet quality benchmarks. It provides the governance structure for Infrastructure, Innovation, and Stakeholder Engagement. In short, it supports the entire SAF value chain, ensuring that every activity is compliant, transparent, and aligned with the mission of producing trustworthy statistics.

VS-08 Innovation and Methodological Development

The Innovation and Methodological Development Value Stream makes sure the statistical organization does not stand still. Its purpose is to discover, test, and introduce new ways of working , new data sources, new methods, new tools, and new technologies. This value stream helps the organization move from “we have always done it this way” to “we can do this better, faster, and smarter.” It connects research, methodology,

IT, and business users. It also makes sure that innovation is controlled , so that only methods that are sound, documented, and compliant end up in production.

PLACEHOLDER: Add ArchiMate view of VS-08

Stage	Name	Value added
VS-08.1	Identify Innovation Needs and Opportunities	Promising developments are known
VS-08.2	Conduct Research and Proof-of-Concepts	Ideas are tested in a limited scope
VS-08.3	Evaluate Methodological Soundness and Feasibility	Only sound and feasible innovations proceed
VS-08.4	Develop and Document New Methodologies or Tools	Innovations are ready for reuse
VS-08.5	Integrate Innovation into Production Environments	Innovations run in production under governance
VS-08.6	Promote Knowledge Sharing and Capability Building	Staff can apply the new methods

The value created here is continuous improvement. Thanks to this stream, the SAF remains future-proof, able to work with new data (IoT, admin, data spaces), new techniques (ML, small-area estimation), and new operating models (secure processing environments, data sharing). How This Supports the Other Value Streams. - It gives Data Acquisition new ways to connect to data spaces and new sources. - It gives Standardization & Integration new mapping or classification techniques. - It gives Editing & Enrichment smarter imputation and outlier detection. - It gives Estimation & Analysis new models and ML-based methods. - It gives Dissemination new channels (APIs, linked data, interactive views).

So, this value stream keeps the entire SAF alive and evolving.

VS-09 Infrastructure and Platform Enablement

The Infrastructure and Platform Enablement Value Stream provides the technical and digital foundation that supports every part of the statistical organization. Where the other value streams focus on data, methods, or governance, this one ensures that all those activities can actually run safely, efficiently, and at scale. It delivers the IT environments, data platforms, tools, and integration services that make statistical production possible, from data collection to dissemination. It ensures that the systems are secure, resilient, and compliant, while also flexible enough to handle new technologies and data sources as the organization modernizes. This value stream enables the organization to work faster, smarter, and with confidence. It transforms infrastructure from a background function into an active enabler of the statistical mission, ensuring that every user, process, and dataset has the tools and environment it needs to perform reliably.

PLACEHOLDER: Add ArchiMate view of VS-09

Stage	Name	Value added
VS-09.1	Define Digital and Technical Requirements	Infrastructure needs follow from business needs
VS-09.2	Design Platform and Infrastructure Architecture	Platforms are designed for storage, processing, integration and secure access
VS-09.3	Provision and Configure Platforms	Environments are available with the required controls
VS-09.4	Enable Integration and Orchestration Services	Processes and services are connected

Stage	Name	Value added
VS-09.5	Ensure Performance, Security and Continuity	Platforms are reliable and protected
VS-09.6	Evolve and Modernise the Technical Ecosystem	The technical landscape stays current

The value created here is operational capability and continuity, a stable, modern, and adaptive foundation for the entire SAF ecosystem.

Connecting to Other Value Streams The Infrastructure and Platform Enablement Value Stream supports every other part of the SAF. It underpins the business and data processes by providing the digital environment in which they operate. - It gives Data Acquisition the secure pipelines and storage needed for data collection. - It gives Standardization, Editing, and Analysis the computing and orchestration needed to process large datasets efficiently. - It gives Dissemination the platforms, portals, and APIs through which users access results. - It ensures Governance and Compliance have the tools to monitor security, quality, and data lineage, and - It enables Innovation by offering scalable, modern platforms for experimentation and research.

VS-10 Customer and Stakeholder Engagement

The Customer and Stakeholder Engagement Value Stream ensures that the statistical organization remains connected, relevant, and responsive to the needs of those it serves. It focuses on building and maintaining mutual understanding and trust between the organization and its users, data providers, partners, and policy stakeholders. In the SAF, this value stream is essential because official statistics exist to create public value. To achieve that, the organization must understand what stakeholders need, communicate transparently, and ensure that products, data services, and policies remain aligned with real-world demands.

This value stream is not a one-way process. It is a continuous dialogue, listening to users, understanding their challenges, and translating those insights into better products, improved services, and stronger relationships.

PLACEHOLDER: Add ArchiMate view of VS-10

Stage	Name	Value added
VS-10.1	Identify and Segment Stakeholders	The NSI knows its users, providers and partners
VS-10.2	Collect and Analyse User Needs	Needs are translated into requirements
VS-10.3	Define and Manage Service Portfolios	Services and products match the needs
VS-10.4	Deliver Services and Manage Relationships	Users and partners are served
VS-10.5	Engage and Communicate with Stakeholders	Stakeholders understand the output
VS-10.6	Monitor Impact and Satisfaction	The NSI knows its impact and acts on it

The value created here is relevance and trust, ensuring that the organization's work is understood, used, and appreciated by its partners, users, and society at large.

Connecting to Other Value Streams The Customer and Stakeholder Engagement Value Stream is the voice and mirror of the statistical organization. - It connects external expectations with internal capabilities, closing the feedback loop across the SAF. - It informs Data Acquisition by revealing what data sources or topics matter most to users. - It influences Product Composition and Dissemination by identifying which outputs and formats users prefer. - It strengthens Governance and Compliance through transparent communication about data ethics, confidentiality, and quality. It supports Innovation and Methodology by gathering ideas and feedback from real users and partners.

Through this value stream, the organization ensures that it doesn't just produce data, it produces public value. It turns statistics into impact by aligning what the organization creates with what stakeholders need.

PLACEHOLDER: Value Stream Template and the application of Value Streams to be added to the toolkit

Resources [T6]

The concept

A **resource** is an asset that an organisation owns or controls. Resources are what the organisation uses to perform its capabilities. A capability without the right resources exists on paper only.

The SAF distinguishes six types of resources:

Type	Examples in an NSI
People and skills	Methodologists, data engineers, subject-matter statisticians, architects
Data and information	Registers, survey data, administrative data deliveries, classifications, metadata
Knowledge	Statistical methods, process documentation, quality frameworks
Technology	Data platforms, processing tools, collection systems, secure processing environments
Financial	Structural budget, project funding, international grants
Relationships and agreements	Data sharing agreements, partnerships with other NSIs, supplier contracts

Three characteristics set resources apart:

- **They are owned or controlled.** A resource is something the NSI has, or has secured through an agreement. Access to administrative data under a provision agreement is a resource.
- **They are limited.** People, money and data are scarce. Resources therefore force choices between capabilities.
- **They are shared.** The same resource often serves several capabilities. A data platform supports collection, processing and dissemination at the same time.

For an NSI, data is a resource in its own right. Registers, long time series and trusted relationships with data providers are often harder to replace than any system.

How the SAF applies resources

Why the SAF uses resources

Resources make the capability model realistic. A maturity assessment that only looks at processes misses the most common bottleneck: not enough people with the right skills, or no access to a needed data source. Linking resources to capabilities shows where a capability depends on a single expert, an outdated platform or an agreement that expires.

How resources fit into an NSI's own architecture

Resources are specific to each NSI. The SAF does not list them. An NSI records the resources behind its key capabilities, notes where resources are missing or at risk, and uses this in its courses of action. Resources also link strategy to the lower layers: technology resources are described in more detail in the application and technology architecture, and data resources in the data architecture.

Relationships with other concepts

Related concept	Relationship	Example
Capability Model	Resources are assigned to capabilities	Staff with methodological expertise are assigned to Methodology (2.2.3)
Courses of Action	Courses of action configure or develop resources	A workforce strategy trains data engineers
Assessment (Motivation)	An assessment can record a resource gap	"Two of our three methodologists for small area estimation retire within two years"
Constraints (Motivation)	Constraints limit which resources can be used	Public cloud may not be used for statistical microdata
Data and technology architecture	Data and technology resources are detailed there	The data platform as a technology resource

Examples (illustrative)

Resource	Type	Supports capabilities	Note
Access to the national population register	Data and information; relationships and agreements	1.2.2 Secondary data collection; 2.1.4 Statistical register management	Depends on a provision agreement that must be renewed
Team of methodologists for estimation	People and skills	1.3.5 Estimating population parameters; 2.2.3 Methodology	Small team; knowledge concentrated in few people
Shared data processing platform	Technology	1.3 Processing; 2.1.1 Data lifecycle management	Shared by several statistical domains

PLACEHOLDER: Resource Template and the application of Resources to be added to the toolkit

Courses of Action [T6] [T7]

The concept

A **course of action** is an approach or plan for configuring capabilities and resources to reach a goal. It answers the question *how will we get there?*

Courses of action range from broad to specific. ArchiMate calls the broad, long-term ones **strategies** and the narrower, shorter-term ones **tactics**. There is no hard line between the two. A strategy is usually broken down into several tactics, and tactics are realised by developing capabilities and resources.

Three characteristics set courses of action apart:

- **They are choices.** A course of action chooses one approach over others. "Collect data web-first" rules out paper-first. A plan without a choice is not a course of action.
- **They serve goals.** Each course of action contributes to one or more goals and outcomes. A course of action that serves no goal needs a justification.
- **They change capabilities and resources.** A course of action builds, strengthens, shares or retires capabilities and resources.

Strategy, architecture and roadmap are related but different. A strategy sets the direction and the main choices. The architecture describes the structure that results. A roadmap sets the order and timing of the changes.

How the SAF applies courses of action

Why the SAF uses courses of action

Goals say where the NSI wants to go. Capabilities say what it must be able to do. Courses of action connect the two: they describe the choices the NSI makes to close the gap. Recording them explicitly prevents a common problem: programmes that each make sense on their own, but pull the organisation in different directions.

How courses of action fit into an NSI's own architecture

Courses of action are specific to each NSI. The SAF does not define them. Most NSIs work with three levels:

1. **Organisation strategy.** The overall direction of the NSI, usually in a multi-annual plan approved by its board or ministry.
2. **Domain strategies.** Strategies for a specific area, such as data, metadata, IT, cloud, security or privacy. Each one refines the organisation strategy for its domain.
3. **Tactics and roadmaps.** Concrete approaches and their order in time, for example in a programme plan.

Each NSI names its own strategies. A small NSI may have one IT strategy that covers applications, infrastructure and cloud. A large NSI may have separate strategies for each. What matters is that the strategies are consistent with each other and traceable to the motivation.

The SAF elements help to make strategies consistent. A domain strategy can be described with the same building blocks used in this framework: the drivers and assessments that make it necessary, the goals and outcomes it serves, the principles it applies, the courses of action it chooses, the capabilities and resources it develops, and the roadmap that sets the order. When all strategies use these building blocks, overlaps and conflicts between them become visible. The toolkit offers templates for this.

Choosing a reference model for the data value chain, such as the Steady-State or the Medallion architecture (see Business Architecture), is itself a course of action.

Relationships with other concepts

Related concept	Relationship	Example
Goals and outcomes (Motivation)	Courses of action serve goals and contribute to outcomes	A shared-platform strategy serves cost efficiency (SAF-G1)
Drivers and assessments (Motivation)	Assessments show why a course of action is needed	An assessment of SAF-D5 leads to a decision to retire legacy platforms
Principles (Motivation)	Courses of action apply principles	An application strategy applies SAF-P01 *Reuse before open source, open source before buying, buying before building*
Capability Model	Courses of action are realised by developing capabilities	A metadata strategy develops Metadata management (2.1.2)
Resources	Courses of action configure resources	A workforce strategy develops data engineering skills
Value Streams	Courses of action improve value streams	A web-first collection strategy improves VS-01
Capability 4.1.2 *Policy and planning*	This capability produces the strategies	Translating the organisation strategy into domain strategies

Examples of strategies (illustrative)

The examples show which domain strategies NSIs often develop. They are not a required set. Each NSI decides which strategies it needs and how to combine them.

Strategy	Typical question it answers	Mainly develops
Data strategy	How do we manage and use data as an asset?	2.1 Data management; VS-01 to VS-06
Metadata strategy	How do we make data findable, understandable and reusable?	2.1.2 Metadata management
IT strategy	What do we reuse, buy or build, and on which platforms?	3.2 Computerisation and automation; VS-09
Cloud strategy	Which workloads may run where, and under which conditions?	3.2.5 IT infrastructure; 3.3 Security and privacy
Information security strategy	How do we protect information and systems?	3.3 Security and privacy; VS-07
Privacy and confidentiality strategy	How do we protect respondents and keep statistical confidentiality?	3.3.1 Privacy management; 1.3.7 Statistical disclosure

Other strategies NSIs develop include strategies for data acquisition, dissemination and open data, artificial intelligence, innovation, workforce and skills, and sourcing and partnerships.

Reference strategies

The following published strategies and guidance documents can serve as reference when an NSI develops its own strategies.

Document	Organisation	Type	Relevant for
Statistics Canada Data Strategy	Statistics Canada	Data strategy of an NSI	Data strategy; shows links to IT, cloud, privacy and metadata strategies
Data Strategy	Office for National Statistics (UK)	Data strategy of an NSI	Data strategy
ABS Data Strategy 2025–2030	Australian Bureau of Statistics	Data strategy of an NSI	Data strategy
Data Strategy of the Secretary-General	United Nations	Data strategy of an international organisation	Data strategy
Cloud for Official Statistics (2024)	UNECE	Guidance	Cloud strategy
Privacy-Enhancing Technologies Guide	UN Committee of Experts on Big Data and Data Science	Guidance	Privacy and confidentiality strategy

PLACEHOLDER: Strategy Template and the application of Courses of Action to be added to the toolkit